CHAPTER 6

Government Actions in Markets

ECON101 - Prince Sultan University

OVERVIEW

Learning objectives

Governments often act to change the prices or quantities set in markets. We judge each action with two tools from Chapter 5: **efficiency** and **fairness**.

1. Describe how rent ceilings create housing shortages.
2. Explain how minimum wage laws create unemployment.
3. Explain the effects of a tax.
4. Explain the effects of production quotas and subsidies.
5. Explain how markets for illegal goods work.

OVERVIEW

Six government actions

Price ceiling

Rent ceiling: illegal to charge **more** than a set price.

Price floor

Minimum wage: illegal to pay **less** than a set price.

Tax

A wedge between the price buyers pay and the price sellers receive.

Production quota

An upper limit on the quantity produced.

Subsidy

A government payment to producers for each unit produced.

Prohibition

Penalties on buyers and/or sellers of a good.

For each action ask: Is the outcome **efficient** (*MSB* = *MSC*, no deadweight loss)? Is it **fair**, by the fair-results or the fair-rules view?

A housing market with a rent ceiling

A **price ceiling** (or **price cap**) makes it illegal to charge a price higher than a specified level. In a housing market it is a **rent ceiling**. - define

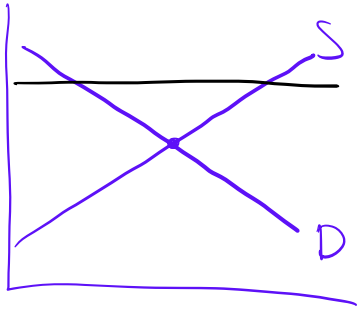
Set above the equilibrium rent

No effect. The market works as if there were no ceiling.

Set below the equilibrium rent

It creates:

1. A housing shortage
2. Increased search activity
3. An illicit market (a black market)



No effect since market can work at the efficient point w/o exceeding limit

Search activity and the illicit market

Increased search activity

Search activity is the time spent looking for someone with whom to do business. It increases when a price is regulated and there is a shortage.

Opportunity cost of housing = **rent** (regulated) + **opportunity cost of search** (unregulated). With fewer apartments, this full cost can exceed the unregulated rent.

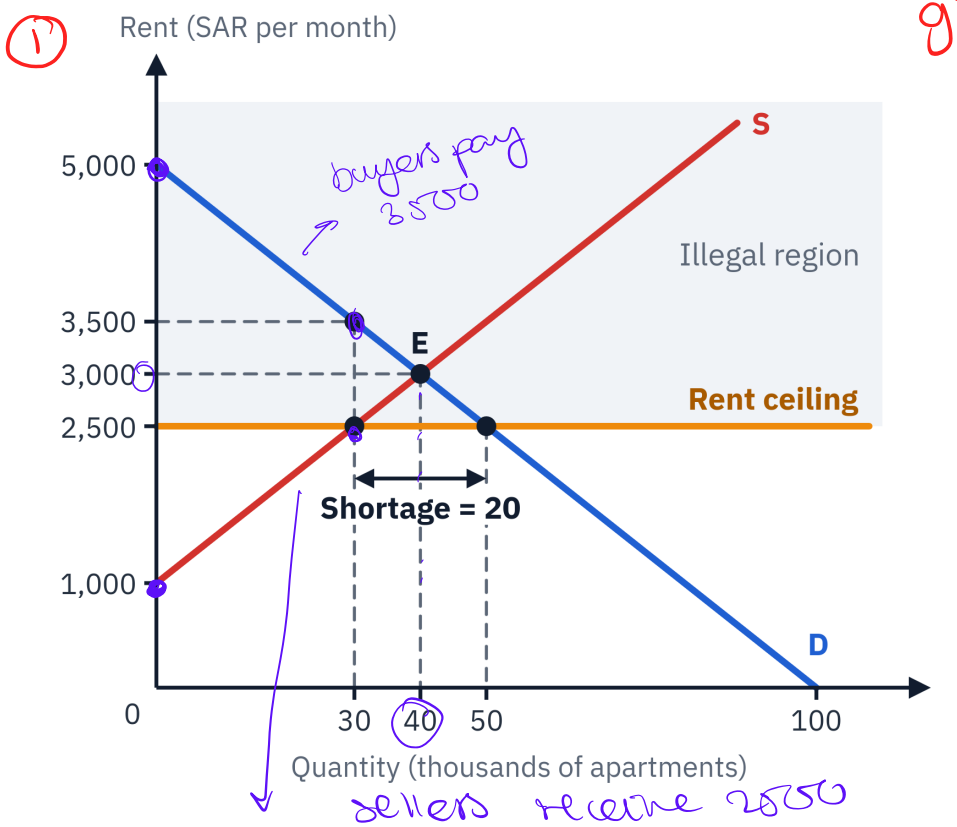
OC = regulated rent + OC of searching
which can be large enough to exceed unregulated rent

An illicit (black) market

Renters and landlords trade illegally at rents above the ceiling. The illicit rent depends on how tightly the ceiling is enforced:

- **Loose enforcement:** close to the unregulated rent.
- **Strict enforcement:** equal to the maximum price a renter is willing to pay for the quantity available.

Example: a rent ceiling creates a shortage



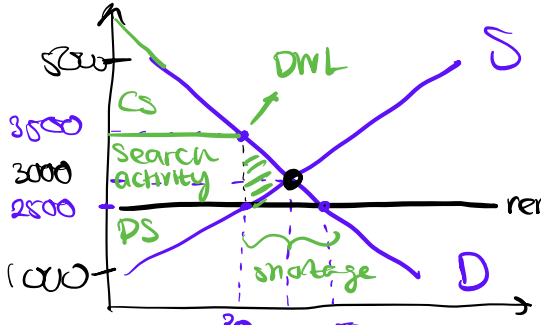
Demand: $P = 5{,}000 - 50Q$
Supply: $P = 1{,}000 + 50Q$

P = rent in SAR per month; Q = thousands of apartments.
Equilibrium: 40 thousand at 3,000 SAR.

With a rent ceiling of **2,500 SAR**:
Demanded = (5,000 − 2,500) ÷ 50 = 50
Supplied = (2,500 − 1,000) ÷ 50 = 30

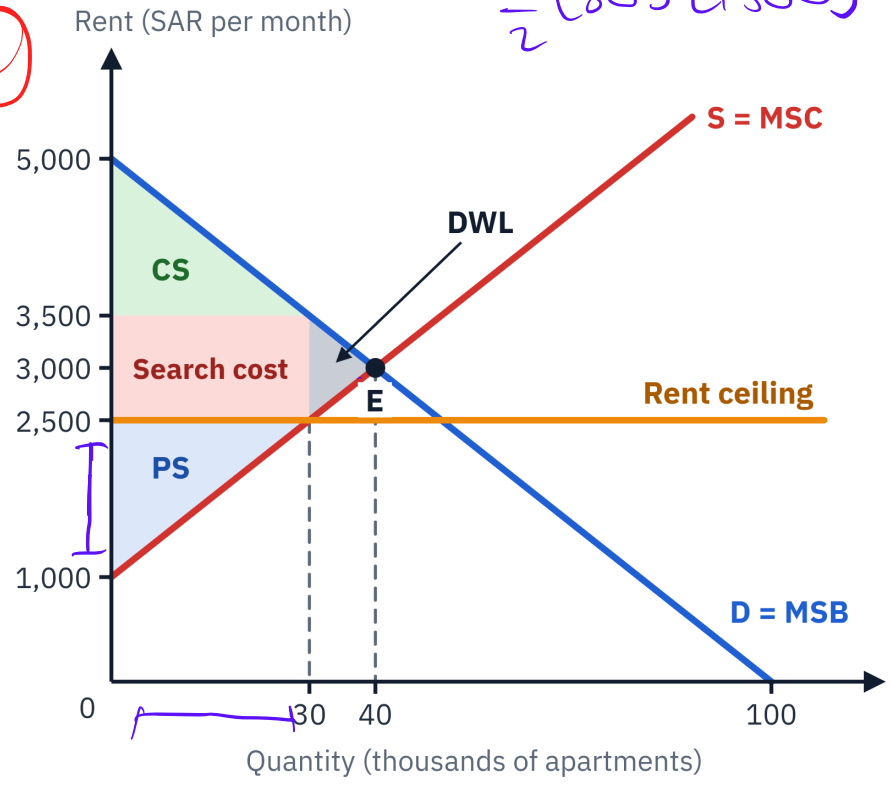
Shortage = 50 − 30 = 20 thousand apartments

Someone will pay 5,000 − 50(30) = **3,500 SAR** for the 30,000th apartment: 2,500 rent + 1,000 of search, or an illicit rent of 3,500.



RENT CEILING

Inefficiency of a rent ceiling



At 30,000 apartments, *MSB* = 3,500 SAR > *MSC* = 2,500 SAR: **underproduction**.

Million SAR per month	No ceiling	Ceiling
Consumer surplus	40.0	22.5
Producer surplus	40.0	22.5
Potential search loss	0.0	30.0
Deadweight loss	0.0	5.0
Total	80.0	80.0

Q in thousands, so DWL = ½ × 10 × 1,000 = 5,000 thousand SAR = 5 million SAR. The table assumes the whole potential search loss occurs.

Full loss = DWL + loss from search

= 5 million SAR

RENT CEILING

Are rent ceilings fair? → theory

Fair-rules view

Unfair: a rent ceiling blocks voluntary exchange.

Fair-results view

Unfair: a rent ceiling does not generally benefit the poor.

The smaller quantity of housing must still be allocated:

Lottery

To the lucky.

First-come, first-served

To those who get on the list first.

Discrimination

By the owner's self-interest.

None of these methods leads to a fair outcome.

MINIMUM WAGE

A labor market with a minimum wage

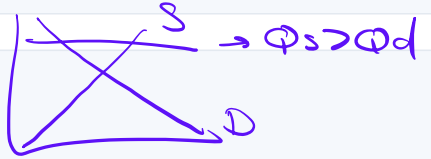
A **price floor** makes it illegal to trade at a price lower than a specified level. In a labor market it is a **minimum wage**.

Set below the equilibrium wage

No effect.

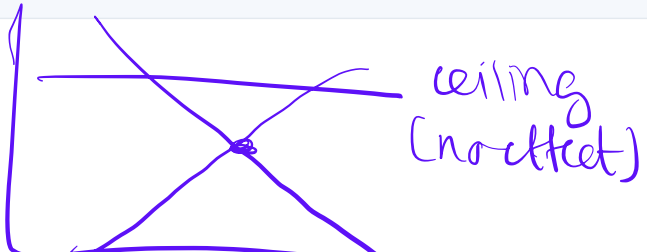
Set above the equilibrium wage

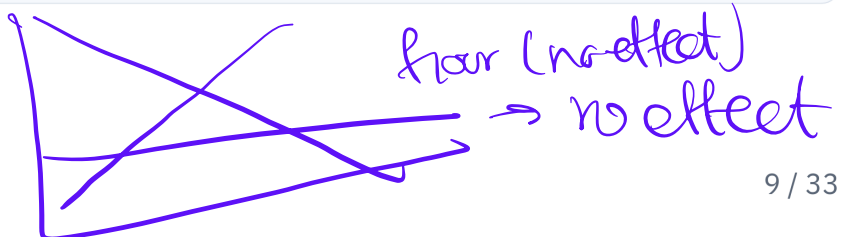
Quantity of labor supplied > quantity demanded. Firms hire only the quantity demanded; the surplus of labor is **unemployment**.



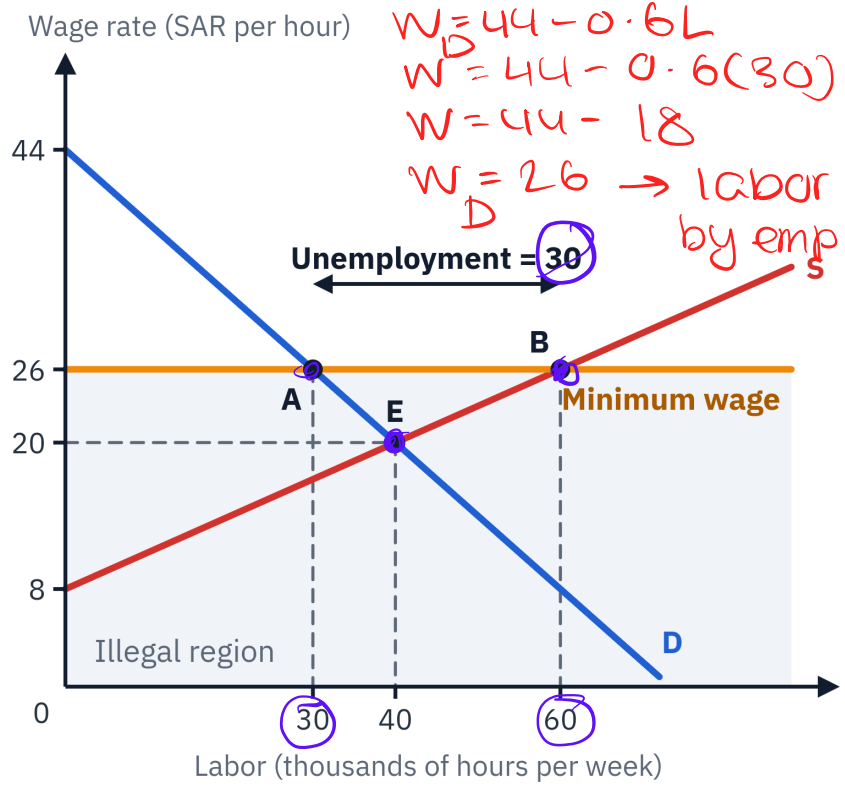
Example: retail assistants

Labor demand $W = 44 - 0.6L$; labor supply $W = 8 + 0.3L$ (W in SAR per hour, L in thousands of hours per week). Equilibrium: 40 thousand hours at 20 SAR. Minimum wage: **26 SAR an hour**.





A minimum wage creates unemployment

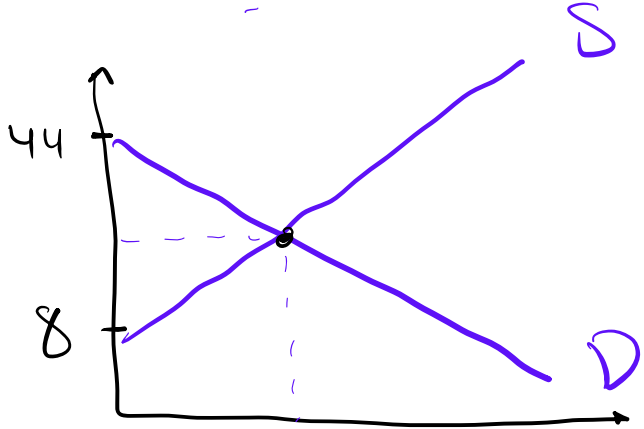


At the minimum wage of 26 SAR:

Quantity demanded = (44 − 26) ÷ 0.6 = 30 (point *A*)
Quantity supplied = (26 − 8) ÷ 0.3 = 60 (point *B*)

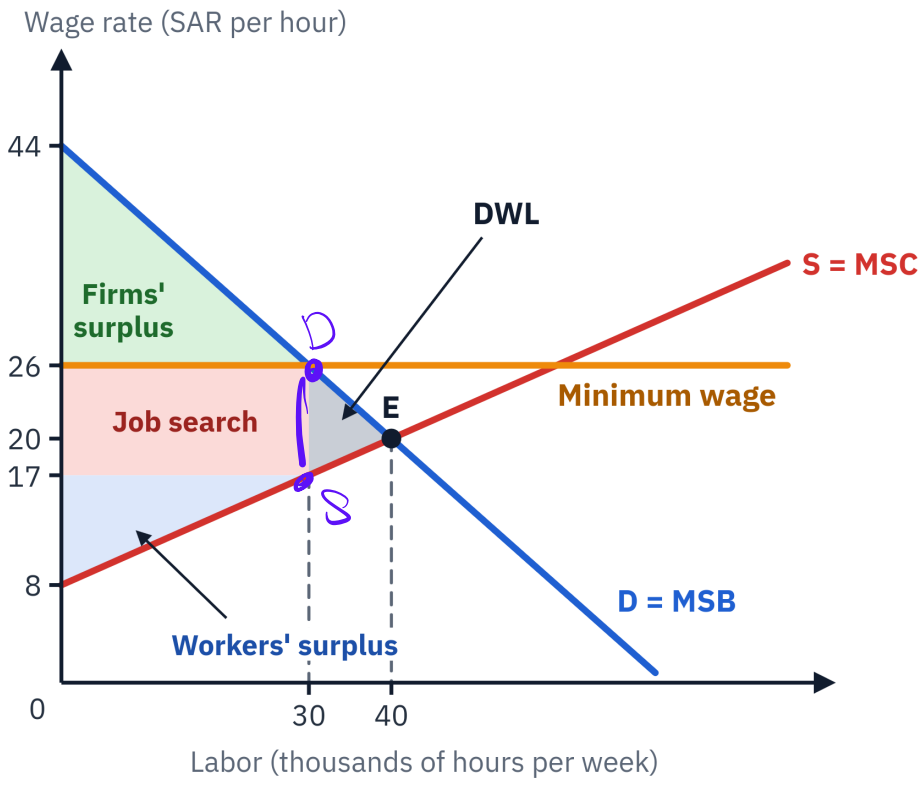
Unemployment = 60 − 30 = 30 thousand hours per week

With only 30 thousand hours demanded, someone is willing to supply the last hour hired for 8 + 0.3(30) = **17 SAR**.



MINIMUM WAGE

Inefficiency of a minimum wage



Supply measures the **marginal social cost** of labor (leisure forgone). Demand measures the **marginal social benefit** (the value of goods and services produced).

- Employment falls to 30, below the efficient 40 thousand hours: *MSB* = 26 > *MSC* = 17.
- DWL = ½ × (40 − 30) × (26 − 17) = **45** thousand SAR per week.
- Potential loss from job search = (26 − 17) × 30 = **270** thousand SAR per week.

Full loss = red area + gray area

MINIMUM WAGE

Is the minimum wage fair?

The minimum wage is unfair on **both** views of fairness.

An unfair result

Only those who have jobs and keep them benefit. The unemployed end up worse off.

An unfair rule

It blocks voluntary exchange. Firms are willing to hire more labor and people are willing to work more, but the law does not permit it.

as a result of the rise of minimum wage

Taxes: who pays?

A tax may be collected from sellers or from buyers, but **the law does not decide who really pays**. **Tax incidence** is the division of the burden of a tax between buyers and sellers.

When an item is taxed, the price buyers pay …

… rises by the full tax

Buyers pay the tax.

… rises by less than the tax

Buyers and sellers **share** the burden.

… does not rise

Sellers pay the tax.

Example: a tax on energy drinks

The market

Suppose a government taxes energy drinks to discourage their consumption.

- Demand: $P = 30 - 0.06Q$
- Supply: $P = 6 + 0.02Q$
- P in SAR per pack; Q in thousands of packs per week
- No tax: $30 - 0.06Q = 6 + 0.02Q$, so $Q = 300$ and $P = 12$ SAR

The tax: 8 SAR per pack

added to supply equation

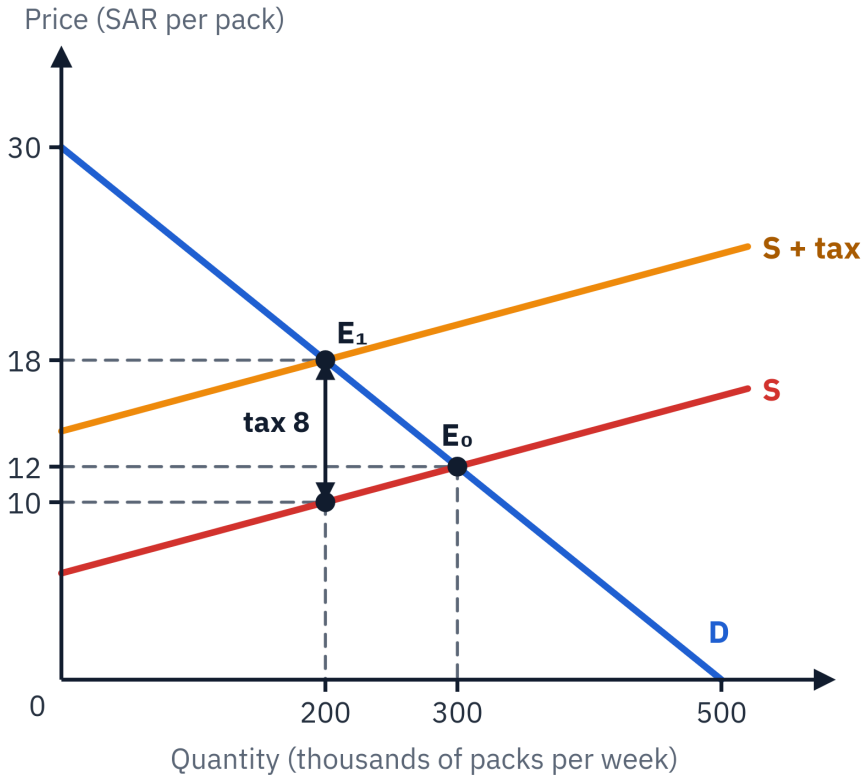
Two prices

The price **buyers pay** includes the tax. The price **sellers receive** excludes it.

Buyers respond to the first and sellers to the second.

Should sellers or buyers pay the tax? The effect is the **same** either way.

A tax on sellers



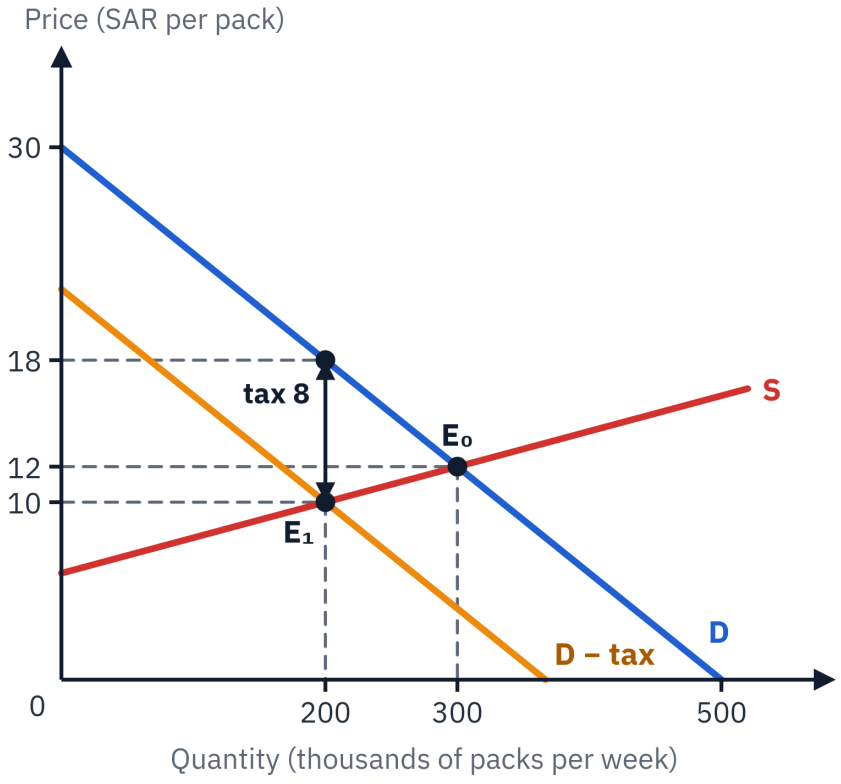
A tax on sellers works like a higher cost, so **supply decreases**. *S* + tax adds 8 SAR to the minimum price sellers accept: $P = 14 + 0.02Q$.

$30 - 0.06Q = 14 + 0.02Q \rightarrow Q = 200, P = 18$

- Quantity falls from 300 to 200 thousand packs.
- Buyers pay **18 SAR**, up 6 SAR.
- Sellers receive 18 − 8 = **10 SAR**, down 2 SAR.

Buyers pay 6 SAR of the tax; sellers pay 2 SAR

A tax on buyers



A tax on buyers lowers what they are willing to pay sellers, so **demand decreases**. D – tax subtracts 8 SAR: $P = 22 - 0.06Q$.

$22 - 0.06Q = 6 + 0.02Q \rightarrow Q = 200, P = 10$

- Sellers receive **10 SAR**.
- Buyers pay 10 + 8 = **18 SAR**.
- Quantity: 200 thousand packs.

The same result as the tax on sellers

$22 - 0.06Q = 6 + 0.02Q$

$+0.06Q \quad +0.06Q$

$22 = 6 + 0.08Q$

$-6 \quad -6$

$16 = 0.08Q$

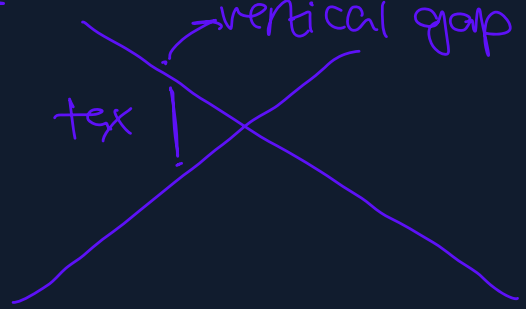
TAXES

The tax as a wedge

A tax drives a **wedge** between the price buyers pay and the price sellers receive. Whichever side is taxed:

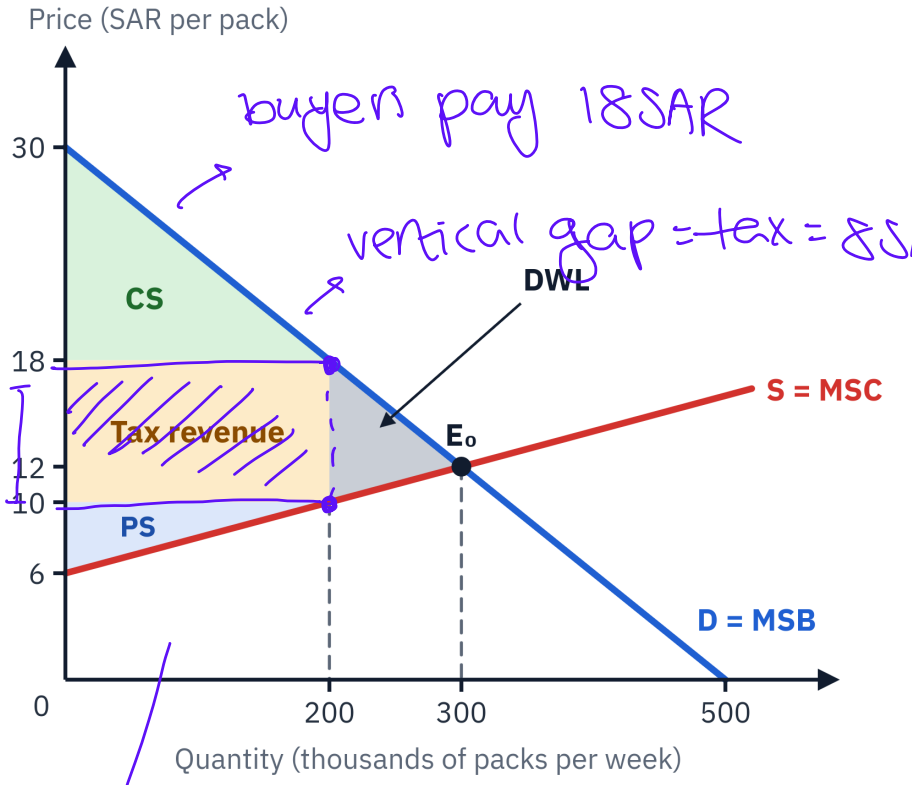
18 SAR	10 SAR	200
the price buyers pay	the price sellers receive	thousand packs per week

With a tax, the equilibrium quantity is not where D and S intersect. It is where the **vertical gap between the curves equals the tax**.



TAXES

Taxes and efficiency



The buyers' price measures *MSB*; the sellers' price measures *MSC*. The wedge makes *MSB* (18) > *MSC* (10).

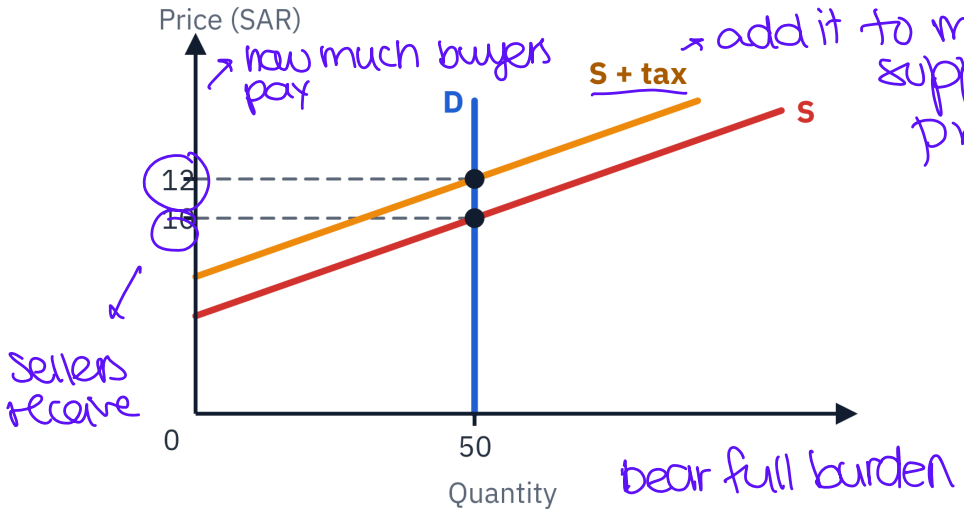
Thousand SAR per week	No tax	Tax of 8
Consumer surplus	2,700	1,200
Producer surplus	900	400
Tax revenue	0	1,600
Deadweight loss	0	400
Total	3,600	3,600

Revenue = 8 × 200 = 1,600. DWL = ½ × (300 − 200) × (18 − 10) = 400.

A tax is inefficient except when demand or supply is **perfectly inelastic**; then the quantity does not change.

Incidence and the elasticity of demand

Each market starts at 10 SAR and 50 units. The tax is 2 SAR per unit.



(a) Perfectly inelastic demand

Quantity stays at 50. Buyers pay 12; sellers still receive 10. **Buyers pay the entire tax**; no deadweight loss.

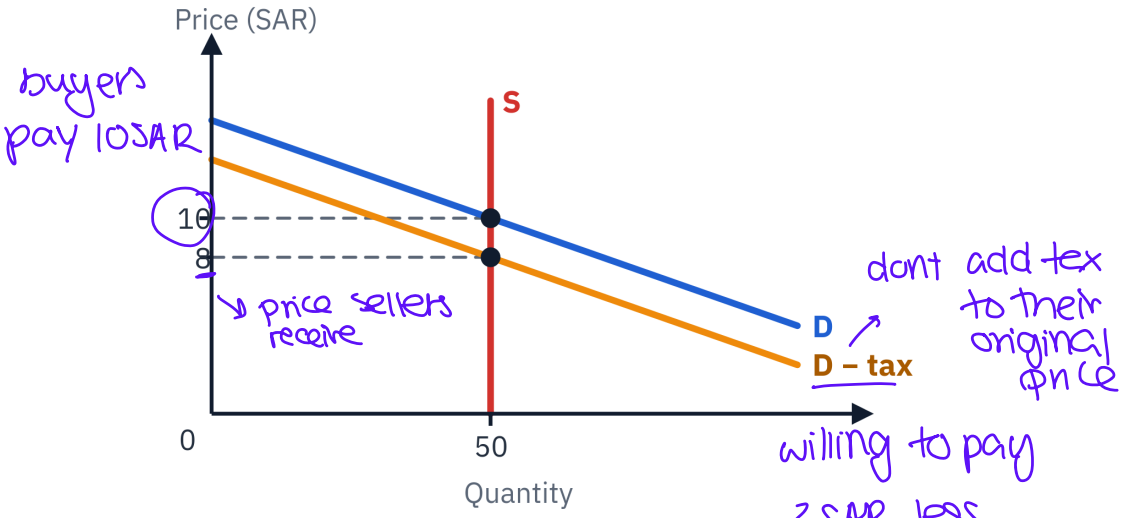


(b) Perfectly elastic demand

Buyers still pay 10; sellers receive 8. Quantity falls to 30. **Sellers pay the entire tax.**

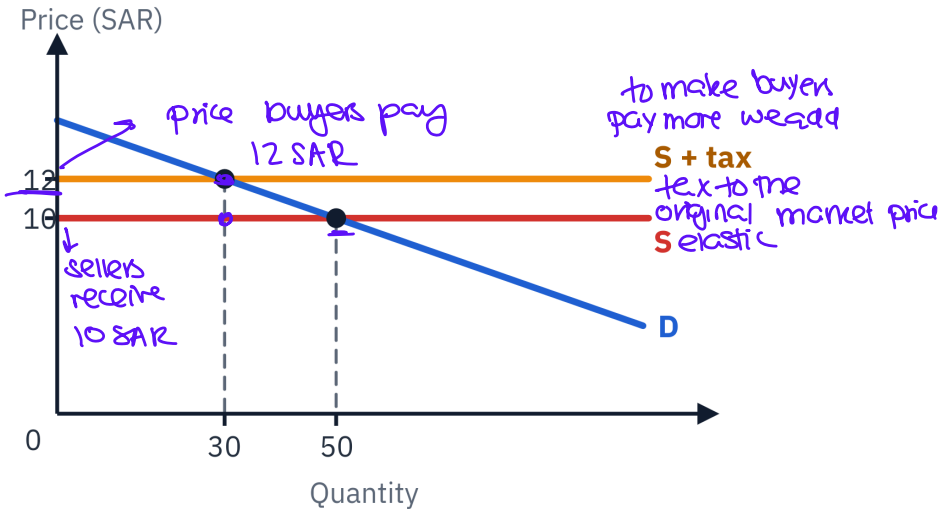
Incidence and the elasticity of supply

Each market starts at 10 SAR and 50 units. The tax is 2 SAR per unit.



(c) Perfectly inelastic supply

Quantity stays at 50. Buyers still pay 10; sellers receive 8.
Sellers pay the entire tax; no deadweight loss.



(d) Perfectly elastic supply

Sellers still receive 10; buyers pay 12. Quantity falls to 30.
Buyers pay the entire tax.

Who pays the tax? A summary

Case (2 SAR tax)	Buyers pay	Sellers receive	Quantity	Tax paid by
(a) Perfectly inelastic demand	12	10	50	Buyers
(b) Perfectly elastic demand	10	8	30	Sellers
(c) Perfectly inelastic supply	10	8	50	Sellers
(d) Perfectly elastic supply	12	10	30	Buyers

The more **inelastic** the demand, the larger is the buyers' share of the tax.

The more **elastic** the supply, the larger is the buyers' share of the tax.

When demand or supply is perfectly inelastic, the quantity does not change and there is no deadweight loss.

Taxes in practice

Taxes are usually levied on goods and services with an **inelastic demand** or an **inelastic supply**.

Tobacco, alcohol and fuel

Inelastic demand, so **buyers** pay most of the taxes on them.

Labor

Low elasticity of supply, so **workers** pay most of the taxes on labor income, such as income and social insurance taxes.

TAXES

Taxes and fairness

Economists propose two conflicting principles of fairness for a tax system:

The benefits principle

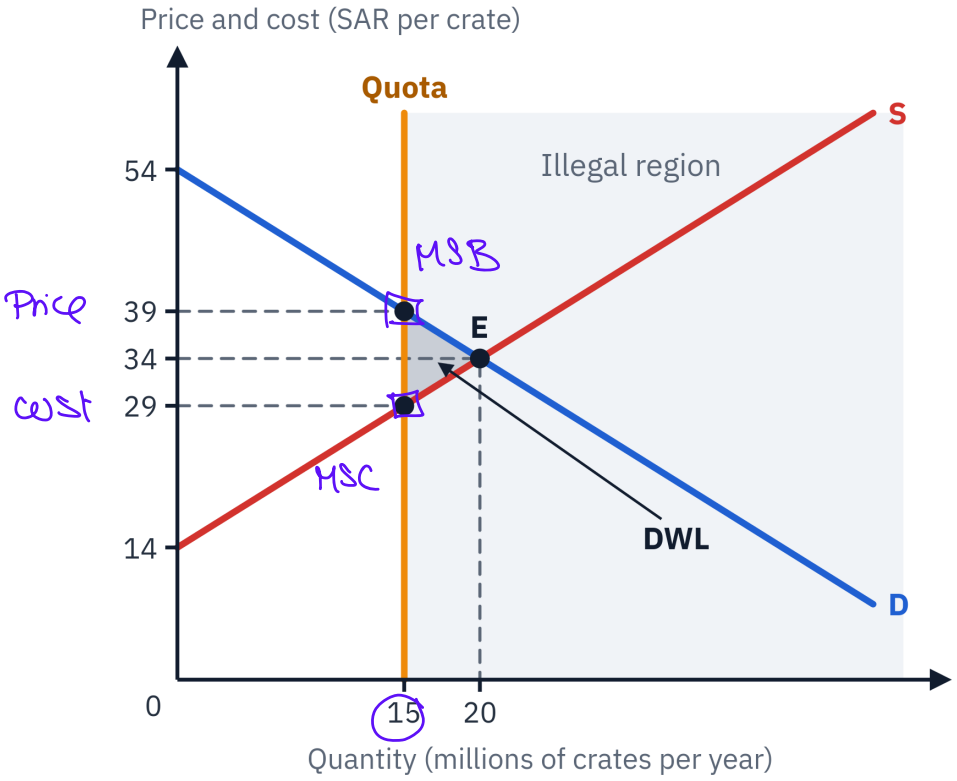
People should pay taxes equal to the benefits they receive from government services. This is fair because those who benefit most pay the most taxes.

The ability-to-pay principle

People should pay taxes according to how easily they can bear the burden. A rich person can bear the burden more easily than a poor person can.

The ability-to-pay principle can reinforce the benefits principle to justify **high rates of income tax on high incomes**.

Production quotas



Governments intervene in farm markets mainly through **production quotas** and **subsidies**.

A **production quota** is an upper limit to the quantity of a good that may be produced in a specified period.

Example: tomatoes

Demand $P = 54 - Q$; supply $P = 14 + Q$ (P in SAR per crate; Q in millions of crates per year).

No quota: $Q = 20$, $P = 34$. Quota: **15 million crates**.

MSB(39) > MSC(29)

deadweight loss = $\frac{1}{2}(10)(5)$

$54 - Q = 14 + Q$

$+Q \quad +Q$

$54 = 14 + 2Q$

$-14 \quad -14$

$\frac{40}{2} = \frac{2Q}{2} \rightarrow Q = 20$

QUOTAS AND SUBSIDIES

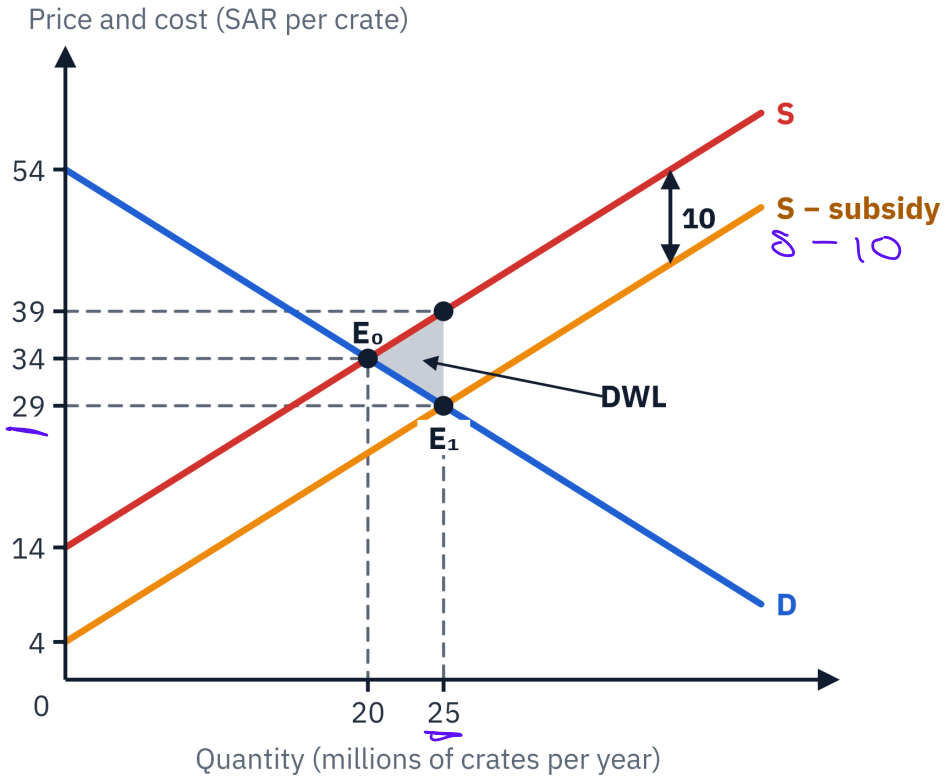
Effects of a production quota

A quota set **below** the equilibrium quantity causes:

Effect	In the example
A decrease in supply	Quantity falls to 15 million crates.
A rise in price	The market price rises to 54 − 15 = 39 SAR.
A decrease in marginal cost	Marginal cost falls to 14 + 15 = 29 SAR.
Inefficient underproduction	MSB 39 > MSC 29; DWL = ½ × 5 × 10 = 25 million SAR per year.
An incentive to cheat	Price exceeds marginal cost, so each farmer gains by producing more than the assigned limit.

↑ exceeding quota limit as a result of rise in price incentivizing them to gain more through producing more than the assigned limit

Subsidies



A **subsidy** is a payment made by the government to a producer. It lowers producers' cost, so **supply increases**.

Example: the same tomato market

The government pays farmers **10 SAR per crate**. The new supply curve is S – subsidy: $P = 4 + Q$.

$54 - Q = 4 + Q \rightarrow Q = 25, P = 29$

Farmers receive 29 + 10 = 39 SAR per crate, equal to the marginal cost of the 25 millionth crate.

Effects of a subsidy

Effect	In the example
A fall in price and an increase in quantity	Price falls to 29 SAR; quantity rises to 25 million crates.
An increase in marginal cost	Farmers receive 29 + 10 = 39 SAR; the marginal cost of the 25 millionth crate rises to 14 + 25 = 39 SAR.
Payments by government → propose paying 10SAR per crate	10 × 25 = 250 million SAR per year.
Inefficient overproduction ↓ exceeds equilibrium	MSC 39 > MSB 29; DWL = ½ × 5 × 10 = 25 million SAR per year. ↓ deadweight loss

Markets for illegal goods

Governments prohibit trade in some goods, such as illegal drugs, yet markets for them still exist. To see how they work, start from a free market and then make trading illegal.

A free market

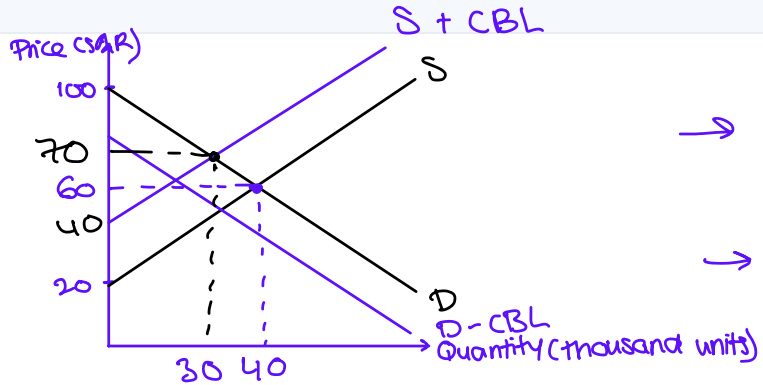
Demand $P = 100 - Q$; supply $P = 20 + Q$ (P in SAR per unit; Q in thousands of units per week).

Equilibrium E: **40 thousand units at 60 SAR**.

The cost of breaking the law (CBL)

The penalties and the chance of being caught.

In this example, **CBL = 20 SAR per unit** for a penalized seller or buyer.



→ If suppliers bear the penalty S+CBL curve shifts to the left/upward

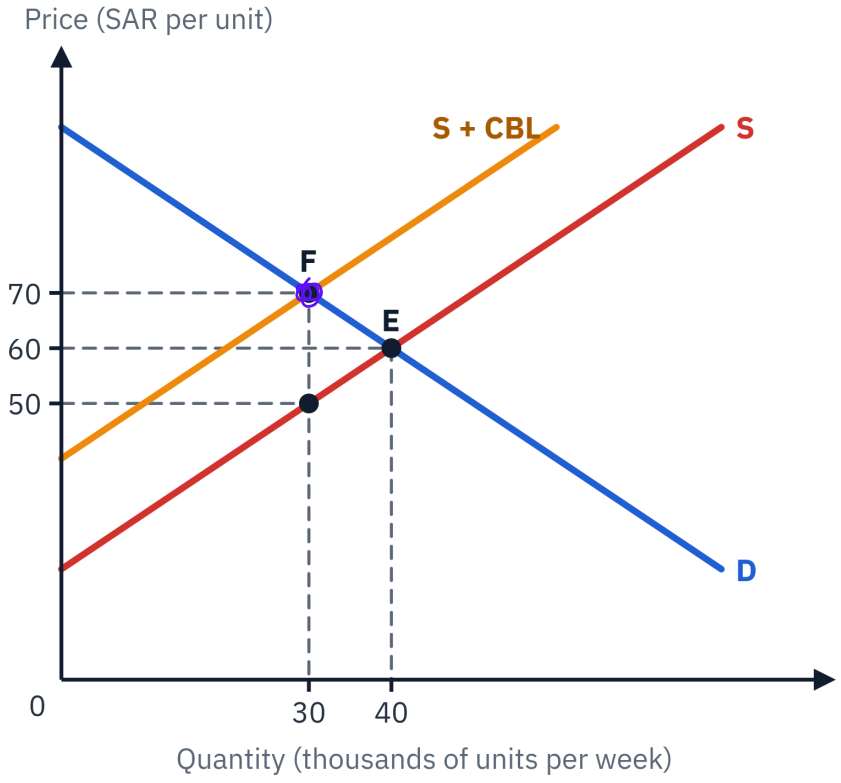
→ if consumers bear the penalty D-CBL curve shifts to the left/downward

ILLEGAL GOODS

Penalties on sellers

why S + CBL → suppliers impose cost of breaking the law into their supply price

D - CBL → reduces what consumers are willing to pay



Sellers add the cost of breaking the law to their minimum supply-price, so **supply decreases** to S + CBL: $P = 40 + Q$.

$100 - Q = 40 + Q \rightarrow Q = 30, P = 70$

demand = S + CBL

- The market price **rises** to 70 SAR (point F).
- The quantity **falls** to 30 thousand units.
- Sellers keep 70 − 20 = 50 SAR.

20 paid as a penalty

initially $D = 100 - Q$

Penalties on buyers

$D - CBL = 80 - Q$



Buyers subtract the cost of breaking the law from what they are willing to pay, so **demand decreases** to D – CBL: $P = 80 - Q$.

new equilibrium

$80 - Q = 20 + Q \rightarrow Q = 30, P = 50$

- The market price **falls** to 50 SAR (point *G*).
- The quantity **falls** to 30 thousand units.
- The buyer's opportunity cost is the market price plus the cost of breaking the law: 50 + 20 = 70 SAR.

after imposing penalties on buyers, their WTP drops to 50

Penalties on both sides



When both sellers and buyers are penalized, **both demand and supply decrease**.

$80 - Q = 40 + Q \rightarrow Q = 20, P = 60$

- The quantity falls to **20 thousand units** (point *H*).
- The market price is 60 SAR.
- The buyer pays 60 + 20 = **80 SAR**.
- The seller receives 60 − 20 = **40 SAR**.

In this example the penalties are equal and *D* and *S* are equally steep, so the market price stays at 60 SAR.

Legalizing and taxing

An illegal good could instead be **legalized and taxed**. A high enough tax would decrease consumption to the level that occurs when trade is illegal.

40 SAR
tax per unit

20
thousand units per week

800
thousand SAR a week in tax revenue

Buyers pay 80 SAR and sellers receive 40 SAR, as at *H*.

The choice between prohibition and taxation also involves arguments that go **beyond economics**.

SOURCE

Source

Michael Parkin (2022) *Microeconomics, Global Edition*, 14th Edition, published by Pearson Education.